

Ryan Liu

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EDUCATION

Northwestern University

B.S. Computer Science and Mathematics | GPA: 3.7/4.00

Expected Graduation: June 2028

Evanston, IL

Relevant Coursework: Graduate Algorithms, Approximation Algorithms, Probability, Operating Systems, Compilers

AWARDS

ICPC North America Championship Honorable Mention

2025, 2026

USA Computing Olympiad Platinum

2025

American Invitational Mathematics Exam (8/15, Top 0.5% of 300,000)

2022, 2023

Outstanding Peer Mentor (Undergraduate Teaching Assistant, Introduction to Algorithms)

2025

EXPERIENCE

Databricks

June 2026 -

Software Engineer Intern

Bellevue, WA

- Productionized rjsonnet, a Rust Jsonnet evaluator/formatter/linter, achieving 2.7x faster formatting and 2.2x faster linting
- Led company rollout to 10% of files, increasing formatter parity from 80% to 99.9% with existing go-jsonnet formatter
- Enabled crucial features in the LSP (e.g. cross-file symbol resolution); Developed an MCP layer exposing the LSP's semantic understanding to coding agents, reducing token usage in simple tasks such as remaining and finding instances; in a demo, reduced token usage by 33x (500 vs 16,000) in a 5-file, 68-instance rename

Amazon

June 2025 – Sep 2025

Software Development Engineer Intern

Seattle, WA

- Developed an end-to-end data pipeline to measure and report call network quality (latency, packet loss, audio jitter), addressing 300,000+ daily missed calls in NA alone and deployed the system to 5 AWS regions worldwide
- Integrated with CloudWatch, SNS, and Lambda to enable real-time alerting across global customer support centers

Northwestern University

September 2024 – Present

Guest Lecturer, Teaching Assistant

Chicago, IL

- Selected to independently prepare and present the Winter 2025 COMP.SCI 212 (Discrete Math) guest lecture to 107 students
- presented on results in number theory, related algorithms, and their applications to encryption

Juxta Technologies (YC S25)

June 2024 – Feb 2025

Founding Software Engineer

Chicago, IL

- Led the development of analytics-driven logistics tools for a \$6M health-tech startup backed by \$600K from Y Combinator, Lynett Capital, Velocity Catalyst, and Authentic Ventures, improving hospital efficiency and reducing operational costs
- Developed algorithms to minimize idle time, optimize task assignment, and streamline workflows using Python, JavaScript, and Firebase, boosting patient transporter efficiency by 44% and reducing transport times by 7.6 minutes per trip

PROJECTS

LC Compiler | COMP.SCI 322

Compilers, C++, Low-level

- Designed and built a multi-pass compiler in C++ for the LC academic programming language, translating LC code through 7 intermediate representations to assembly with semantics-preserving transformations at each stage
- Implemented graph-coloring register allocation, closure convention and heap allocation, AST tiling optimizations

Matching Engine | Personal Project | [github](#) | 40 hours

C++, Low Latency, Concurrency

- Built a C++20 matching engine with single-writer order books, symbol-based parallelism, and price-time priority
- Removed global lock contention and redundant task state; achieved 2x benchmark throughput and 56% p99 latency reduction

Offline Judge | Personal Project | [github](#) | 15 hours

HTTP Requests, Webscraping, C++, Python

- Developed an offline judge for olympiad problems, with test case scraping and runtime/memory limit enforcement
- Designed an system using g++, subprocess, and ulimit, ensuring accurate performance benchmarking

SKILLS

Programming Languages: Python, C++, JavaScript, TypeScript, Java, SQL, MATLAB, R

Softwares and Tools: Git, Bash, React, React Native, Next.js, Node.js, HTML, CSS, NumPy, scikit-learn, OpenCV, REST APIs, GraphQL, Google Cloud Platform, AWS, Jupyter, PostgreSQL, MongoDB, Docker, CI/CD pipelines